UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 10 Physics Final Exam Practice

Term 1 2026–2027 • Set C • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. Which surface most nearly produces specular reflection? [4 marks]

A. Crumpled cloth
B. A smooth polished mirror
C. Rough paper
D. A painted rough wall

Reference: pp. 4–5 | **Answer B**

2. If an object moves 10 cm toward a fixed plane mirror, what happens to its image? [4 marks]

A. It becomes real
B. It moves 10 cm toward the mirror on the other side
C. It moves 20 cm away from the mirror
D. It stays fixed

Reference: pp. 5–8; p. 8, Check Your Progress 8,10 | **Answer B**

3. An object is exactly at C of a concave mirror. Its image is approximately: [4 marks]

A. At F, upright and reduced
B. At infinity
C. Behind the mirror, enlarged
D. At C, inverted and equal in size

Reference: pp. 11–13 | **Answer D**

4. A concave mirror has focal length 7.0 cm. An object is 16.0 cm away. Find X_i. [4 marks]

A. +12.4 cm
B. −12.4 cm
C. +4.9 cm
D. +23.0 cm

Reference: p. 16, Practice Problem 15 | **Answer A**

5. Why can a real mirror image be caught on a screen? [4 marks]

A. Only the backward extensions converge
B. Reflected rays actually converge at the image
C. The image must be upright
D. A screen turns a virtual image into a real one

Reference: pp. 11–13; p. 18 | **Answer B**

6. A ray enters an unknown material from air at 30.0° and refracts at 20.0°. What is the material's index? [4 marks]

A. 1.00
B. 0.684
C. 2.00
D. 1.46

Reference: p. 25, Check Your Progress 36 | **Answer D**

7. For glass (n = 1.50) to air, the critical angle is about 41.8°. Which incidence gives TIR? [4 marks]

A. 0°
B. 30°
C. 40°
D. 50°

Reference: pp. 23–25 | **Answer D**

8. What is dispersion? [2 marks]

A. Reflection from a perfectly smooth surface
B. The complete absorption of white light
C. The merging of all colours into one wavelength
D. Separation of light into component wavelengths

Reference: pp. 24–25 | **Answer D**

9. Why is the colour order reversed in a secondary rainbow? [4 marks]

A. The droplets are all opaque

B. Light undergoes two internal reflections in the droplets

C. The Sun changes the order of its colours

D. Only red light is refracted

Reference: pp. 24–25 | **Answer B**

10. Which image magnification is possible for a real object with a single concave lens? [4 marks]

A. +0.40

B. +2.5

C. −0.40

D. −2.5

Reference: p. 35, Check Your Progress 54 | **Answer A**

11. Spherical aberration occurs because: [4 marks]

A. All rays meet at exactly one point

B. The lens is necessarily opaque

C. Edge rays and near-axis rays focus at different positions

D. Only violet light is transmitted

Reference: p. 31 | **Answer C**

12. An uncorrected myopic eye focuses distant light: [4 marks]

A. Behind the retina

B. At the cornea with zero refraction

C. In front of the retina

D. Only on the retina

Reference: pp. 32–33 | **Answer C**

13. Why does monochromatic light form alternating bright and dark double-slit fringes? [4 marks]

A. The slits switch on and off alternately

B. Only one slit emits light at a time

C. The path difference varies across the screen

D. Gravity sorts the photons

Reference: pp. 42–44 | **Answer C**

14. Why can a soap film show different colours in different places? [4 marks]

A. Each place contains a different coloured lamp

B. The speed of light in vacuum varies across the film

C. Its thickness varies and changes the interference conditions

D. The film converts all light into sound

Reference: pp. 46–47 | **Answer C**

15. What causes the spectrum seen on a compact disc under white light? [4 marks]

A. Diffraction and interference from its repeated tracks

B. Radioactive decay of the coating

C. Absorption of every wavelength except one

D. A separate light source in each track

Reference: pp. 53–54 | **Answer A**

16. Why are different wavelengths seen at different angles after a diffraction grating? [2 marks]

A. The constructive-interference angle depends on wavelength

B. Every wavelength has a different speed in vacuum

C. Only one groove reflects at a time

D. The screen changes the frequency of light

Reference: pp. 53–55 | **Answer A**

Set C Section B Free Response

17. A light ray reflects from a plane mirror at 35° to the normal. Also consider the convex mirror below. [10 marks]

Reference: p. 6, Practice Problem 2; pp. 9–14

(a) Find the angle of incidence and the angle the incident ray makes with the mirror surface. [2]

Answer Incidence 35° [1]; angle with surface 90° − 35° = 55° [1].

(b) Draw a labelled reflection sketch showing the normal and both angles. [3]

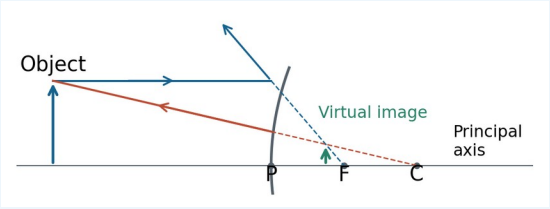
Answer Mirror and perpendicular normal [1]; incoming/outgoing arrows [1]; equal 35° angles measured from normal [1].

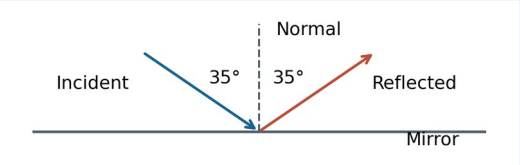
(c) Complete the convex-mirror ray diagram and state the nature of its image. [3]

Answer Two correctly reflected rays with dashed backward extensions [2]; virtual, upright, diminished image behind mirror [1].

(d) Why is a convex mirror useful for observing traffic? Where is its principal focus? [2]

Answer Wide field of view [1]; virtual focus behind mirror on principal axis [1].





18. Light passes from water (n = 1.33) at 31° to the normal into a plastic block, where it refracts at 27°. A separate diverging lens has f = −5.0 cm and a 6.5 cm tall object at X_o = 6.0 cm. [10 marks]

Reference: p. 21, Practice Problem 33; p. 35, Check Your Progress 56

(a) Calculate the block's refractive index. [3]

Answer $n_{water}\sin31° = n_{block}\sin27°$ [1]; $n_{block} = 1.33\sin31°/\sin27°$ [1] = 1.51 [1].

(b) Calculate the critical angle from this block to air. [2]

Answer $\sin\theta_c = 1/1.51$ [1]; $\theta_c \approx 41.5°$ [1].

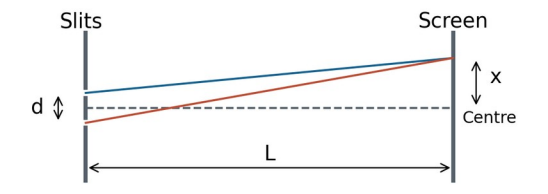
(c) Find X_i using the thin-lens equation. [3]

Answer $1/f = 1/X_o + 1/X_i$ [1]; $X_i = (−5)(6)/(6 + 5)$ [1] = −2.73 cm [1].

(d) Find magnification and image height; state whether the image is real or virtual. [2]

Answer $m = −X_i/X_o = +0.455$ [1]; $h_i = +2.95$ cm, upright and virtual [1].

19. A double-slit investigation uses light of wavelength 632.8 nm. The screen is 1.000 m from the slits and the first bright band is 65.5 mm from the central band. [10 marks]



Reference: p. 45, Practice Problem 3; pp. 42–44; p. 49, Check Your Progress 13

(a) On the sketch, identify d, L and x, and state the condition for a first-order bright band. [3]

Answer d: slit separation; L: screen distance; x: distance from central band [2 for all three, 1 for two]. Path difference = λ [1].

(b) Calculate the slit separation using the small-angle approximation. [3]

Answer $d = \lambda L/x$ [1]; $d = (632.8 \times 10^{-9})(1.000)/0.0655$ [1] = 9.66×10^{-6} m [1].

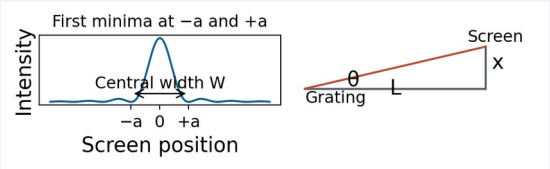
(c) Find the distance of the second-order bright band from the central band. [2]

Answer $x_2 = 2x_1$ [1] = 131.0 mm [1].

(d) Describe the central fringe and nearby fringes if white light replaces the single wavelength. [2]

Answer Central fringe is white [1]; nearby fringes are coloured, with violet closer to centre than red [1].

20. Yellow light passes through a 0.0295 mm single slit. The central bright band is 24.0 mm wide on a screen 0.600 m away. Separately, a DVD has groove spacing 741 nm. Blue light gives a first-order spot 0.580 m from the centre on a wall 0.650 m away. [10 marks]



Reference: p. 53, Practice Problem 18; p. 56, Example Problem 3 and Practice Problems 20, 23

(a) Calculate the yellow light's wavelength. [3]

Answer $\lambda = Ww/(2L)$ [1]; $\lambda = (0.0240)(0.0295 \times 10^{-3})/(2 \times 0.600)$ [1] = 590 nm [1].

(b) How would a longer wavelength affect the central width, with slit and screen unchanged? [1]

Answer It would increase the width, since W is proportional to λ [1].

(c) Find the blue light's first-order diffraction angle. [2]

Answer $\theta = \tan^{-1}(0.580/0.650)$ [1] = 41.7° [1].

(d) Calculate the blue wavelength and describe a white-light grating pattern. [4]

Answer $\lambda = d\sin\theta$ [1] = $741\sin41.7° \approx 493$ nm [1]. White central maximum [1]; separated coloured spectra on both sides [1].